Software Engineering

Lab 7

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Section 5-J

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Prefix Run

```
PS C:\Vedant\PESU\Semester5\SE\Lab\Lab7> python electricity_bill.py
==== Running units_consumed ====
Enter meter readings as previous:current (e.g. 1200:1350): 1200:1350
Units Consumed: 150

==== Running daily_average ====
Enter daily units as CSV (e.g. 5,7,6): 5,7,6
Daily Average: 6.0

==== Running tariff_rate ====
Enter a tariff code (e.g. DOM-A): DOM-A
Rate per Unit: 3
PS C:\Vedant\PESU\Semester5\SE\Lab\Lab7>
```

Hypothesis Errors

```
PS C:\Vedant\PESU\Semester5\SE\Lab\Lab7> python -m pytest test_electricity_bill.py -v
============================= test session starts =============================
platform win32 -- Python 3.12.6, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\vshar\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Vedant\PESU\Semester5\SE\Lab\Lab7
plugins: hypothesis-6.170.14
collected 3 items

test_electricity_bill.py::test_units_consumed_no_crash FAILED            [ 33%]
test_electricity_bill.py::test_daily_average_no_crash FAILED             [ 66%]
test_electricity_bill.py::test_tariff_rate_no_crash FAILED               [100%]
```

```
================================================================================ FAILURES ================================================================================
_____________________________________________ test_units_consumed_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\test_electricity_bill.py", line 5, in test_units_consumed_no_crash
  |     def test_units_consumed_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\test_electricity_bill.py", line 6, in test_units_consumed_no_crash
    |     units_consumed(s)
    |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\electricity_bill.py", line 6, in units_consumed
    |     return int(current) - int(previous)
    |            ^^^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_units_consumed_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\test_electricity_bill.py", line 6, in test_units_consumed_no_crash
    |     units_consumed(s)
    |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\electricity_bill.py", line 5, in units_consumed
    |     previous, current = reading.split(':')
    |     ^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    | Failing test case: test_units_consumed_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\test_electricity_bill.py", line 6, in test_units_consumed_no_crash
    |     units_consumed(s)
    |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\electricity_bill.py", line 5, in units_consumed
    |     previous, current = reading.split(':')
    |                         ^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_units_consumed_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________________________ test_daily_average_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\test_electricity_bill.py", line 9, in test_daily_average_no_crash
  |     def test_daily_average_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\test_electricity_bill.py", line 10, in test_daily_average_no_crash
    |     daily_average(s)
    |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\electricity_bill.py", line 9, in daily_average
    |     units = [int(u) for u in units_csv.split(',')]
    |              ^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_daily_average_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\test_electricity_bill.py", line 10, in test_daily_average_no_crash
    |     daily_average(s)
    |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\electricity_bill.py", line 9, in daily_average
    |     units = [int(u) for u in units_csv.split(',')]
    |                              ^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_daily_average_no_crash(
    |     s=None,
    | )
    +------------------------------------
```

```
_____ test_tariff_rate_no_crash _____
+ Exception Group Traceback (most recent call last):
|   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\test_electricity_bill.py", line 13, in test_tariff_rate_no_crash
|     def test_tariff_rate_no_crash(s):
|                    ^^^
| ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\test_electricity_bill.py", line 14, in test_tariff_rate_no_crash
  |     tariff_rate(s)
  |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\electricity_bill.py", line 14, in tariff_rate
  |     return RATES[category.upper()] * SLAB[slab.upper()]
  |            ~~~~~^^^^^^^^^^^^^^^^^^
  | KeyError: ''
  | Failing test case: test_tariff_rate_no_crash(
  |     s='-',
  | )
  +---------------- 2 ----------------
  | Traceback (most recent call last):
  |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\test_electricity_bill.py", line 14, in test_tariff_rate_no_crash
  |     tariff_rate(s)
  |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\electricity_bill.py", line 13, in tariff_rate
  |     category, slab = code.split('-')
  |     ^^^^^^^^^^^^^^
  | ValueError: not enough values to unpack (expected 2, got 1)
  | Failing test case: test_tariff_rate_no_crash(
  |     s='',  # or any other generated value
  | )
  +---------------- 3 ----------------
  | Traceback (most recent call last):
  |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\test_electricity_bill.py", line 14, in test_tariff_rate_no_crash
  |     tariff_rate(s)
  |   File "C:\Vedant\PESU\Semester5\SE\Lab\Lab7\electricity_bill.py", line 13, in tariff_rate
  |     category, slab = code.split('-')
  |                      ^^^^^^^^^^
  | AttributeError: 'NoneType' object has no attribute 'split'
  | Failing test case: test_tariff_rate_no_crash(
  |     s=None,  # or any other generated value
  | )
  +------------------------------------
```

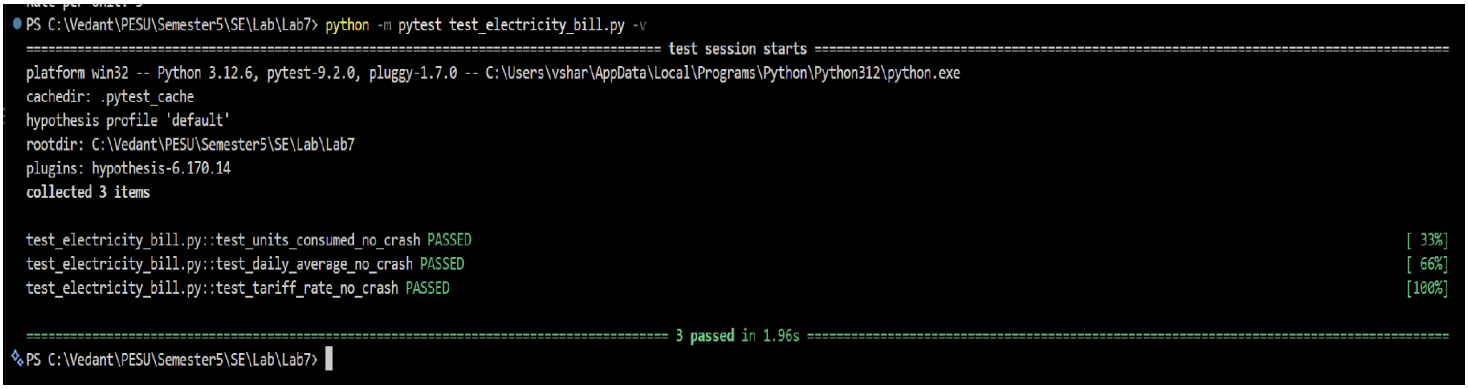
```
=========================== short test summary info ============================
FAILED test_electricity_bill.py::test_units_consumed_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_electricity_bill.py::test_daily_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_electricity_bill.py::test_tariff_rate_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================== 3 failed in 3.14s ===============================
```

Postfix

```
PS C:\Vedant\PESU\Semester5\SE\Lab\Lab7> python electricity_bill.py
==== Running units_consumed ====
Enter meter readings as previous:current (e.g. 1200:1350): 1200:1350
Units Consumed: 150

==== Running daily_average ====
Enter daily units as CSV (e.g. 5,7,6): 5,7,6
Daily Average: 6.0

==== Running tariff_rate ====
Enter a tariff code (e.g. DOM-A): DOM-A
Rate per Unit: 3
```



test_electricity_bill.py

```
from hypothesis import given, strategies as st

from electricity_bill import units_consumed, daily_average, tariff_rate



@given(st.one_of(st.text(), st.none()))

def test_units_consumed_no_crash(s):

    units_consumed(s)



@given(st.one_of(st.text(), st.none()))

def test_daily_average_no_crash(s):

    daily_average(s)



@given(st.one_of(st.text(), st.none()))

def test_tariff_rate_no_crash(s):

    tariff_rate(s)
```